Evidence Links of the Training Project — LavaRápido (Vehicle Washing)

Web and mobile booking and management platform for a car wash

Field	Value
Program / group	Technologist in Software Analysis and Development (ADSO), group 3145556
Center	Centro de la Industria, la Empresa y los Servicios — SENA, Huila Region
Training project	Vehicle Washing Management System (Vehicle Washing / LavaRápido), as it appears on the cover of the deliverables. To be confirmed with the instructor whether the ficha records it under another name (owner: Brayan Cabrera)
Work team	Vehicle Washing Development Team (team 8 in the instructor's evaluation of 10/21/2026)
Members	Juan Camilo Guilombo Osorio, Brayan Estiven Patiño Cabrera, Merari Urbano Quintero
Instructor	Karol Daniela Correa
Delivery date	Final presentation planned for the week of October 27, 2026; the instructor confirms the exact date (owner of confirming it: Brayan Cabrera)
Last update of this document	October 27, 2026

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1. Evidence Matrix

Documentation repositories used in the matrix

- *`lavarapido-docs`* (public, `main` branch): https://github.com/BrayanCABRERA123/lavarapido-docs
- *`vehicle-w-docs`* (private, `code-sena` organization, `docs` branch): https://github.com/code-sena/vehicle-w-docs/tree/docs. It is the source technical documentation of the project (SDD); the documents in `lavarapido-docs` are derived from it. Each evidence links the equivalent public section first, so there is always an accessible link.

Structure of `lavarapido-docs`: the general index (README) links every evidence and the change history (CHANGELOG) records each version. Each evidence has a numbered folder (`01_` to `11_`) and, inside it, `ES/` and `EN/`, each with `MD/` (Markdown and its `capturas/` screenshots), `PDF/`, and `WORD/`; the article (10) is bilingual in a single file, and this matrix (11) has Markdown and PDF. The matrix links the English Markdown version; the PDF and Word files are in the neighboring folders.

#	Evidence	Minimum verifiable content	Direct link	Access status	Remarks
1	Requirements specification report	Functional and non-functional requirements, user stories, acceptance criteria, traceability	SRS v8.0 (EN) · Functional requirements (SRS §5) · Non-functional requirements (SRS §6) · User stories per requirement (SRS §10) · Source in `vehicle-w-docs`: Functional requirements · Non-functional requirements · User stories	Public / Pending permissions (`vehicle-w-docs`)	The SRS follows IEEE 830 and ISO/IEC/IEEE 29148 and covers 41 functional requirements (RF-001 to RF-038) and use cases CU01 to CU32. Spanish version: `01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS/ES/`.
2	Software analysis	Dynamic views, processes, use cases, sequence and state diagrams	Dynamic views analysis (EN) · Sequence, activity, and state diagrams · Source in `vehicle-w-docs`: UML diagrams (sources)	Public / Pending permissions (`vehicle-w-docs`)	The screenshots are in `02_DOCUMENTACION_DE_ANALISIS_DEL_SOFTWARE/EN/MD/capturas/`. The diagrams include the booking, payment, registration, reminder, assignment, and inspection sequences, and the booking and payment states.
3	Technical proposal of the software	Selected alternative, technologies, justification, scope, and constraints	Technical proposal (EN) · Design decisions (SRS §3.6) · Alternatives considered · Source in `vehicle-w-docs`: Architecture decisions (ADR-001 to ADR-015)	Public / Pending permissions (`vehicle-w-docs`)	The ADRs justify the microservices architecture, the database, the API Gateway, JWT, and the use of .NET for the payment service.
4	Software design	Graphical interfaces, architecture, patterns, data model, and diagrams	Software design (EN) · Architectural model · Data model · API and system routes (technical manual §13) · Source in `vehicle-w-docs`: Architecture · Data model · OpenAPI contracts	Public / Pending permissions (`vehicle-w-docs`)	Hexagonal architecture pattern (ADR-007). Section 2.5 describes the responsive design. The screenshots are in `04_DOCUMENTACION_DE_DISENO_DEL_SOFTWARE/EN/MD/capturas/`.
5	Source code and software artifacts	Repositories of the web frontend, mobile app, backend, infrastructure, and database, with their branches	Code artifacts (EN) · see section 2	Public	The integration branch is `develop`. `main` is behind in every repository (see section 2).

#	Evidence	Minimum verifiable content	Direct link	Access status	Remarks
6	Test documentation	Plan, test cases, results, execution evidence, defects, and fixes	Test documentation (EN) · E2E tests · Test plan · Source in `vehicle-w-docs`: Testing strategy	Public / Pending permissions (`vehicle-w-docs`)	Follows ISO/IEC/IEEE 29119-3. Automated (689) and end-to-end (19) tests run on 10/21/2026; functional (42), performance, and responsive design tests on 10/19/2026 (§3). The traceability matrix is in §5 and the findings in §6.
7	Implementation procedure	Reproducible installation, configuration, deployment, and verification	Implementation procedure (EN) · Azure deployment document (EN) · Support: Infrastructure README · Team guide	Public	Available in Markdown, PDF, and Word (ES and EN) in `07_PROCEDIMIENTO_DE_IMPLEMENTACION/`. The deployment document records the deployment of 10/16/2026 on Azure with its evidence screenshots.
8	Corrective, preventive, and improvement actions	Findings, priority, owner, applied action, and closing evidence	Corrective, preventive, and improvement actions (EN) · Support: Preventive actions · Traceability between actions, risks, and technical debt · Source in `vehicle-w-docs`: Risks · Technical backlog	Public / Pending permissions (`vehicle-w-docs`)	Available in Markdown, PDF, and Word (ES and EN) in `08_ACCIONES_CORRECTIVAS_PREVENTIVAS_Y_DE_MEJORA/`.
9	Technical, installation, and user manuals	Prerequisites, configuration, use, and troubleshooting	Technical manual (EN) · Installation manual (EN) · User manual (EN) · Customer user manual (EN) · Operator user manual (EN) · Administrator manual (EN) · PDF of the manuals · Source in `vehicle-w-docs`: User manual	Public / Pending permissions (`vehicle-w-docs`)	Each manual is in Markdown, PDF, and Word (ES and EN), in the `09_MANUALES/` folder. Six manuals: technical, installation, administrator, user (customer and operator together) and, since 10/26/2026, one user manual per role (customer and operator, version 1.0), as the SENA manuals guide recommends. All have real screenshots of the system; the `.env` secrets appear masked.
10	Project article	Problem, objectives, solution, methodology, results, and conclusions	APA 7 article (ES/EN) · APA 7 PDF · IEEE version · ACM version	Public	The three versions (APA 7, IEEE, and ACM) are in Markdown, PDF, and Word in the `10_ARTICULO_DEL_PROYECTO/` folder. Each file is bilingual (ES and EN in the same document), which is why this folder is not split by language.
11	Deployed application or demo	Access URL or demo environment and verification instructions	Application in production · Android app download · Deployment document (EN)	Public	Deployed on 10/16/2026 on an Azure virtual machine (Docker Compose, Caddy with Let's Encrypt HTTPS) and last updated on 10/22/2026 with the changes of 10/21/2026 (migrations 112, 040, and 041). The machine is turned on on demand: if the URL does not answer, it is off. The smoke tests and screenshots are in section 5 of the deployment document. The credentials are delivered through a private channel (see section 7).
12	Traceability and integration of evidence	Relationship between requirements, user stories, modules, tests, and versions	Traceability matrix (SRS §10) · Requirements–tests traceability (§5) · Source in `vehicle-w-docs`: Traceability matrix	Public / Pending permissions (`vehicle-w-docs`)	There are no release tags, so traceability to versions is done by branch (`develop`) and by commit.

2. Source Code Repositories

Status as of 10/27/2026, taken from the repositories: on 10/27/2026 *`v1.1.2`* of the web, the mobile app, payment-service, and infra and *`v1.1.1`* of customer-service and booking-service shipped, and payment-service *`v1.1.3`* was approved (PR #16, pending publication: that is why its `develop` is ahead). on 10/26/2026 *`v1.1.0`* shipped in the 10 repositories (the mobile app got its first version) and that same day *`v1.1.1`* in the web, the mobile app, payment-service, and infra (terms and conditions in the third person with the inspection report section, test counts, and payment-service deployed from `main`). In all of them, `main` is the latest tag and `develop` has no pending code changes (the "`develop` ahead" column only counts commits that are not in `main`). The previous version, `v1.0.0`, is from 10/23/2026 (10/24/2026 in payment-service).

Component	Technology	Repository	Version	`main`	`develop` (last commit)	`develop` ahead of `main`	Access
Web frontend	Angular	Front-end-proyecto-web	[`v1.1.2`] (https://github.com/BrayanCABRERA123/Front-end-proyecto-web/tree/v1.1.2)	`acea5e3` (= `v1.1.2`)	`f25d407`	0 commits (`main` includes it)	Public
Mobile app	React Native (Expo)	LavadoVehicular-Movil	[`v1.1.2`] (https://github.com/merari0322/LavadoVehicular-Movil/tree/v1.1.2)	`f8300d6` (= `v1.1.2`)	`f8300d6`	0 commits (`main` includes it)	Public
API Gateway	Spring Boot	lavarapido-api-gateway	[`v1.1.0`] (https://github.com/BrayanCABRERA123/lavarapido-api-gateway/tree/v1.1.0)	`6961feb` (= `v1.1.0`)	`97dbf39`	0 commits (`main` includes it)	Public
Security (authentication, JWT, roles)	Spring Boot	lavarapido-security-service	[`v1.1.0`] (https://github.com/BrayanCABRERA123/lavarapido-security-service/tree/v1.1.0)	`391fb1f` (= `v1.1.0`)	`0a664a9`	0 commits (`main` includes it)	Public
Customers and vehicles	Spring Boot	lavarapido-customer-service	[`v1.1.1`] (https://github.com/camiloguilombo12/lavarapido-customer-service/tree/v1.1.1)	`83d322e` (= `v1.1.1`)	`83d322e`	0 commits (`main` includes it)	Public
Catalog and bookings	Spring Boot	lavarapido-booking-service	[`v1.1.1`] (https://github.com/camiloguilombo12/lavarapido-booking-service/tree/v1.1.1)	`506feca` (= `v1.1.1`)	`506feca`	0 commits (`main` includes it)	Public
Operations (operators and bays)	Spring Boot	lavarapido-operation-service	[`v1.1.0`] (https://github.com/BrayanCABRERA123/lavarapido-operation-service/tree/v1.1.0)	`9511328` (= `v1.1.0`)	`15ff0f4`	0 commits (`main` includes it)	Public
Payments and loyalty	.NET Core	lavarapido-payment-service	[`v1.1.2`](https://github.com/merari0322/lavarapido-payment-service/tree/v1.1.2)	`42dac3d` (= `v1.1.2`)	`4ccc52e`	10 commits (`v1.1.3` approved in PR #16, pending publication)	Public
Notifications	Spring Boot	lavarapido-notification-service	[`v1.1.0`] (https://github.com/BrayanCABRERA123/lavarapido-notification-service/tree/v1.1.0)	`7fdfd1e` (= `v1.1.0`)	`fd8eec7`	0 commits (`main` includes it)	Public
Infrastructure (Docker, database, E2E tests)	Docker, SQL Server, RabbitMQ	lavarapido-infra	[`v1.1.2`] (https://github.com/BrayanCABRERA123/lavarapido-infra/tree/v1.1.2)	`682f629` (= `v1.1.2`)	`682f629`	0 commits (`main` includes it)	Public
Technical documentation (SDD)	Markdown, OpenAPI	vehicle-w-docs (`docs` branch)	—	483 commits behind `docs`	`ba7dde6` (`docs` branch, 10/27/2026)	—	Pending permission
Documentary evidence	Markdown	lavarapido-docs (`main` branch)	—	working branch	—	—	Public

Database: a single `LavaRapido` database on SQL Server 2022 (Express in production and Developer locally), with one schema per service (ADR-003, ADR-009). Each Spring Boot service manages its migrations with Liquibase. The payment service migrations are in [`lavarapido-payment-service/db`](https://github.com/merari0322/lavarapido-payment-service/tree/develop/db).

> Recommendation: tell the evaluator that the delivered version is the latest tag of each repository on `main` (`v1.1.1` in the web, the mobile app, payment-service, and infra, and `v1.1.0` in the others); `develop` has the same code.

3. Deployment and Technical Verification

Aspect	Value
Version (release tag)	`v1.1.2` of the web, the mobile app, payment-service, and infra and `v1.1.1` of customer-service and booking-service (10/27/2026); `v1.1.0` in api-gateway, security, operations, and notification (10/26/2026). payment-service `v1.1.3` approved and pending publication (section 2). The first version, `v1.0.0`, is from 10/23/2026
Environment	Academic production on Azure (Ubuntu 24.04 VM, Docker Compose with `docker-compose.prod.yml`), deployed on 10/16/2026 and last updated on 10/24/2026 with that day's `develop` code (two runs over SSH; 11 healthy containers, migrations 044, 045, 113, 047, 116, 022, and 023 applied, and 0 errors in the logs), with `:prev` images to go back with `rollback.sh` and the protection that prevents downgrading. Until `v1.1.0` (10/26/2026) the server was updated with `BRANCH=develop`; since `lavarapido-infra` `v1.1.1`, `setup-server.sh` pulls `main` in every repository, payment-service included. For development: local with Docker (`docker compose --profile app up -d --build` from `lavarapido-infra`). Details in the deployment document.
Public demo environment	https://lavarapidovehicular.northcentralus.cloudapp.azure.com
Continuous deployment	Active since 10/26/2026: with the secrets created, merging "Release v1.1.1" into `main` of `lavarapido-infra` deployed on its own at 22:19 (Colombia time) and the requests from the other repositories kept coming until 22:48, all green (healthy services and web and API at 200); payment-service is also deployed from `main`. In the mobile app, APK `versionCode` 6 and the over-the-air update were published after checking that the native fingerprint matched. Details in the deployment document, sections 1.3.12 and 4.3.4

3.1 Role Matrix

Role	Main functions	Client where it is used
Administrator	Users and roles, catalog, promotions, schedules and bays, operators, bookings, inspection report, payments, reports	Web and mobile
Customer	Registration, vehicles, bookings, payments by QR or transfer, history, ratings, notifications	Web and mobile
Operator	Assigned services, schedule, execution, history, ratings received	Web and mobile

3.2 Health Endpoints (Verified on 10/16/2026 in the Local Environment)

Service	Port	Endpoint	Result
API Gateway	8080	`/actuator/health`	`UP`
security-service	3001	`/actuator/health`	`UP`
customer-service	3002	`/actuator/health`	`UP`
booking-service	3003	`/actuator/health`	`UP`
operations-service	3004	`/actuator/health`	`UP`
payment-service	3005	`/health`	`ok` (checks SQL Server; answers 503 when the database does not respond)
notification-service	3006	`/actuator/health`	`UP`

3.3 Interactive API Documentation (Swagger/OpenAPI)

- **Spring Boot services (ports 3001–3004 and 3006):** `http://localhost:<port>/swagger-ui/index.html` answered `200` on 10/16/2026.
- **payment-service (.NET Core):** Swagger is only enabled in the `Development` environment. Under Docker it is not exposed (`404`). To see it, run the service with `ASPNETCORE_ENVIRONMENT=Development`.
- **Versioned contracts:** [`07-api/contracts/openapi`](https://github.com/code-sena/vehicle-w-docs/tree/docs/07-api/contracts/openapi) in `vehicle-w-docs` (private). Public version: API and system routes, in the technical manual.

3.4 Smoke Tests

Test	Procedure	Result
Authentication	`POST http://localhost:8080/api/v1/auth/login` with a test user and check that it returns a JWT	200 with `accessToken` (10/16/2026, test account of the *dev* profile)
Read	With the JWT, query the service catalog or the customer's bookings	`GET /api/v1/catalog/services` and `GET /api/v1/bookings/me` answered 200 (10/16/2026)
Write	Create a booking and check that it appears in the history and that the notification arrives	Covered by `full-flow.test.js`: booking created, automatically assigned to the chosen operator, payment approved, service started and finished, rating, and every notice in the customer inbox (10/16/2026)
Automated E2E flow	`node --test "e2e/*.test.js"` in `lavarapido-infra` (with `E2E_OPERATOR_ID=2` in the local database)	19 of 19 tests passed in 11.2 s (10/19/2026); on 10/21/2026 all 19 passed again in the early morning and in the evening, with the gateway session check on

3.5 Evidence Screenshots

Screenshot	Description	Link
Web screens (login, admin, customer, operator)	28 interface screenshots	[SRS `capturas` folder](https://github.com/BrayanCABRERA123/lavarapido-docs/tree/main/01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS/EN/MD/capturas)
Dynamic views and modals	41 screenshots	[Analysis `capturas` folder](https://github.com/BrayanCABRERA123/lavarapido-docs/tree/main/02_DOCUMENTACION_DE_ANALISIS_DEL_SOFTWARE/EN/MD/capturas)
Design (landing, login, booking, payments)	Screenshots of the final design	[`capturas` folder](https://github.com/BrayanCABRERA123/lavarapido-docs/tree/main/04_DOCUMENTACION_DE_DISENO_DEL_SOFTWARE/EN/MD/capturas)
Real screenshots of the manuals	110 screenshots per language of the running system (10/16, 10/19, 10/26, and 10/27/2026): screens of the three roles on the web and of the customer and operator in the mobile app (the customer and operator ones also with the `customer-*` and `operator-*` names of the role manuals), installation terminal, and data model diagram	[Manuals `capturas` folder](https://github.com/BrayanCABRERA123/lavarapido-docs/tree/main/09_MANUALES/EN/MD/capturas)
Containers, logs, health checks, and migrations in production	Portainer, startup logs, `/actuator/health`, clean console, and Liquibase changesets (10/16/2026)	[Deployment `capturas` folder](https://github.com/BrayanCABRERA123/lavarapido-docs/tree/main/07_PROCEDIMIENTO_DE_IMPLEMENTACION/EN/MD/capturas)

4. Accessibility and Quality Checklist

Status as of 10/23/2026; unchecked boxes have their owner.

- [ ] Every link opens correctly, without 404 errors or denied access, tested from an account outside the team. *To be tested with an outside account (Brayan Cabrera). The internal links of the 31 Markdown files were reviewed on 10/23/2026 and the only broken one, in the Administrator Manual, was fixed.*
- [ ] The instructor has read permission on `code-sena/vehicle-w-docs` (private) and checked it with their account. *Pending (Brayan Cabrera); also, `main` of that repository is 366 commits behind `docs`.*
- [ ] Each evidence in `lavarapido-docs` has its Markdown, PDF, and Word in ES and EN, and the PDF and Word files open correctly. *The Markdown files are current as of 10/24/2026; the PDF and Word files of the documents changed on October 23 and 24, and those of the technical proposal (03), are still to be regenerated with the team's generator.*
- [x] The documents are readable: the Markdown renders on GitHub and the PDF files open without errors. *Checked on the version of 10/21/2026; repeat after regenerating.*
- [ ] Each repository identifies the project (README), the evaluated branch (`develop`), and the version (release tag). *`v1.0.0` in 9 of 10 repositories (payment-service since 10/24/2026); the mobile app is missing, with PR #1 open (Merari Urbano).*
- [ ] The documentation, repositories, and application are consistent: what the SRS and the manuals describe exists in the `develop` code. *Partial: SRS v8.0 marks 3 RF and 9 RNF as partial and the Corrective Actions document records what is still open (16 actions, none P0 or P1).*
- [ ] No simulations (mocks, `json-server`, sample data) are presented as real functionality. *Since 10/24/2026 the app and the web no longer show the business sample data (seed 113 removes it from the database); the app's legal texts still say "Express Car Wash S.A.S." (team decision).*
- [ ] There are no credentials or sensitive data: `.env` is not versioned (only `.env.example`) and there are no tokens or passwords in the history. *Partial: see the security criterion in section 5.*
- [ ] The exported PDF keeps the links active (click at least one per section). *Pending when the PDF is regenerated (Brayan Cabrera).*

5. Acceptance Criteria

Criterion	Fulfillment condition	Result / remarks
Completeness	The 12 evidences have a direct and verifiable link	The 12 evidences have a link. Evidence 11 links the application deployed on Azure since 10/16/2026.
Accessibility	Every link opens for the instructor without asking for permissions	Partial: the links of `lavarapido-docs` and the code repositories are public (checked without logging in); those of `vehicle-w-docs` require giving the instructor read permission.
Traceability	Requirements → user stories → modules → tests → versions related	There is a matrix in §10 of the SRS (v8.0, with the partial requirements marked), another in §5 of the test document, and the source matrix in `vehicle-w-docs`. Version `v1.0.0` exists in 9 of 10 repositories.
Consistency	The documentation matches the code and the running application	Latest `develop` CI of 10/24/2026: 1,119 automated tests and 73 in the mobile app (1,192), all passed; that same code runs on the public server since that day. On 10/23/2026 there were 838. Tests of 10/21/2026 (final run, in the evening): 707 automated and 19 end-to-end tests passed; the early-morning run had 689. That day the administrator's "To close" list was also checked live on the web and in the mobile app. Local environment verified on 10/19/2026: every service `UP`, 42 functional cases passed, responsive design verified on 66 web screens and 15 app screens, and the screenshots of the manuals, SRS, analysis, and design taken again that day.
Security	No secrets or personal data exposed	Partial (reviewed on 10/24/2026 and updated on 10/26/2026): no repository versions `.env` and these documents contain no passwords. In `develop` the development passwords were removed from the code, the compose file, the READMEs and, since 10/24/2026, from `GUIA-EQUIPO.md` and the `payment-migrate` arguments, but these remain: the development `sa` password in `.env.example` of the `v1.0.0` tag of `lavarapido-infra` (since `v1.1.0`, `main` has `<cambia-esta-contraseña>`), the old SQL Server password in payment-service's Git history (its `main` reads it from `DB_PASSWORD`), and the Firebase key in the app's `google-services.json` (action ACO-25). The screenshots use test data.
Presentation	Navigable document, readable tables, and PDF with active links	Pending: the team regenerates the PDF with its APA 7 generator and checks one link per section (owner: Brayan Cabrera).

6. Delivery

Format	Suggested name
Markdown	`11_ENLACES_DE_EVIDENCIAS/EN/MD/Evidence_Links_Project_VehicleWashing.md` (in Spanish: `ES/MD/Enlaces_Evidencias_Proyecto_VehicleWashing.md`)
PDF	`11_ENLACES_DE_EVIDENCIAS/EN/PDF/Evidence_Links_Project_VehicleWashing.pdf`; to deliver it, copy it with the date, for example `Evidence_Links_Project_VehicleWashing_2026-10-17.pdf`

The delivery channel (platform, email, shared folder, or other) is set by the instructor.

7. Security Note

- Do not include in this document or in the repositories passwords, tokens, API keys, the content of `.env`, connection strings, or real personal data.
- **Test credentials:** if the evaluator needs to log in, the team will deliver through a private channel (direct message or institutional platform) one test user per role (administrator, customer, and operator). Those accounts are created only for the evaluation and are deactivated afterwards.
- Do not use links to local files (`C:\...`, `file://`), expired links, or links that require unresolved permissions.
- **Public repositories:** review the history (`.env` files, JWT secrets, database passwords) before publishing or delivering them.
- **Private repositories** (`vehicle-w-docs`): give the instructor read permission and check the access with their account.

8. Reference Resources

- GitHub Docs: About repositories
- GitHub Docs: Setting repository visibility

9. Reminder for the Team

> Update this matrix every time locations, branches, versions, or permissions change. After each change, export the PDF again and check the links again in the exported PDF.